

AV NETWORK DESIGN REPORT

Example Hotel Ballroom AV

October 1, 2026

Americas · Catalog 2026.09.27 · Request R-EXAMPLE

VALIDATED BY THE PRO AV DESIGN TEAM

Reviewed by Pro AV Design team, September 30, 2026

82

DEVICES

8

SWITCHES

1,326 W

POE LOAD

2.6 kW

EST. POWER

11.0 U

RACK SPACE

Networking engineered for AV over IP

NETGEAR AV switches, routers and WiFi access points are built specifically for networked media and AV, with out-of-the-box performance and easy setup for even the most complex installation. NETGEAR works with AV manufacturers and integration partners around the world, and supports the partners and associations that make the Pro AV and broadcast industries work.

● Optimized for AV over IP

Low-latency switching for audio, video and control, with PoE+ (30 W) and Ultra PoE++ (90 W) per port options.

● Certified AV profiles

NETGEAR AV OS with configuration profiles certified by leading AV manufacturers (Audinate, Crestron, Q-SYS, SMPTE ST 2110, NDI).

● Multicast that just works

NETGEAR IGMP Plus stops multicast flooding automatically, and Auto-LAG and Auto-Trunk connect switches without manual setup.

● Free central management

The NETGEAR Engage Controller manages switches, routers and WiFi 7 access points from one screen, and can set up the network before the hardware arrives.

● Scales with the project

From a single rack to redundant, active-active cores with 10G, 25G and 100G uplinks.

● Lower total cost

Energy-efficient hardware, a lifetime warranty on managed switches and less need for specialist IT resources.

IN THIS DESIGN

This design uses M4250 and M4350 switches, WBE758 WiFi 7 access points and the PR460X Pro Router, all managed for free from one screen in the NETGEAR Engage Controller with NETGEAR AV OS.

BUILT FOR EVERY SPACE

Broadcast and live events · Corporate and commercial AV · Higher education · Medical · Unified communications · High-end residential

HELP FROM DESIGN TO DEPLOYMENT

NETGEAR has a dedicated Pro AV and broadcast network design team, pre-sales support and technical assistance. Free on-demand training and certification at academy.netgear.com covers networking basics through in-depth troubleshooting, and the top courses are AVIXA-accredited for CTS renewal units.

Support and services

Every NETGEAR managed switch includes a lifetime hardware warranty and complimentary support to get you started. Premium Support tiers add guaranteed response times, longer support hours and faster hardware replacement, so you can match support to how critical the system is.

	Warranty (included)	Sprint	Overdrive	Fastlane
Complimentary setup support	✓	✓	✓	✓
AI-driven chat support	✓	✓	✓	✓
Direct access to support engineers		✓	✓	✓
Support hours		24/5	24/7	24/7
Guaranteed response time (P1)		8 hours	2 hours	Custom
Priority case handling			✓	✓
Escalation to senior engineers			Level 2	Level 3
Hardware replacement	Fast RMA	Advance replacement	Next business day	Custom
Scheduled support calls			✓	✓
Operational reports			Quarterly	Custom
Designated or dedicated engineer				✓
Custom workflows and SLAs				✓

Managed and enterprise products include 90 days of complimentary phone, chat and online support. Fully managed switches include 3 years of Sprint Support. Sprint and Overdrive response times are engagement commitments for Priority 1 cases.

PROFESSIONAL SERVICES

Deployment services	Design validation, hands-on configuration and readiness testing, so the network is production-ready from day one. Remote, onsite or hybrid.
Health checks and optimization	Proactive network audits that find issues before they become outages, with a clear remediation plan.
Custom training and enablement	Training based on your deployed system: architecture walkthroughs, feature deep dives, best practices and troubleshooting.
Interoperability, staging and migration	Multi-vendor compatibility checks, staged configurations and low-risk migrations to NETGEAR.
Event services	Pre-event validation and real-time onsite or remote engineering support for live broadcasts, sporting events and other high-stakes shows.
Fastlane professional engagement	A high-touch engagement with senior engineering access for organizations that need deeper collaboration.
Part number	Service
PSP1104-10000S	On-site engineering assistance, full day
PRC0001-10000S	New remote configuration, full day

Pricing depends on project scope and the level of support, and includes travel and expenses. Ask your NETGEAR contact or distributor for a quote.

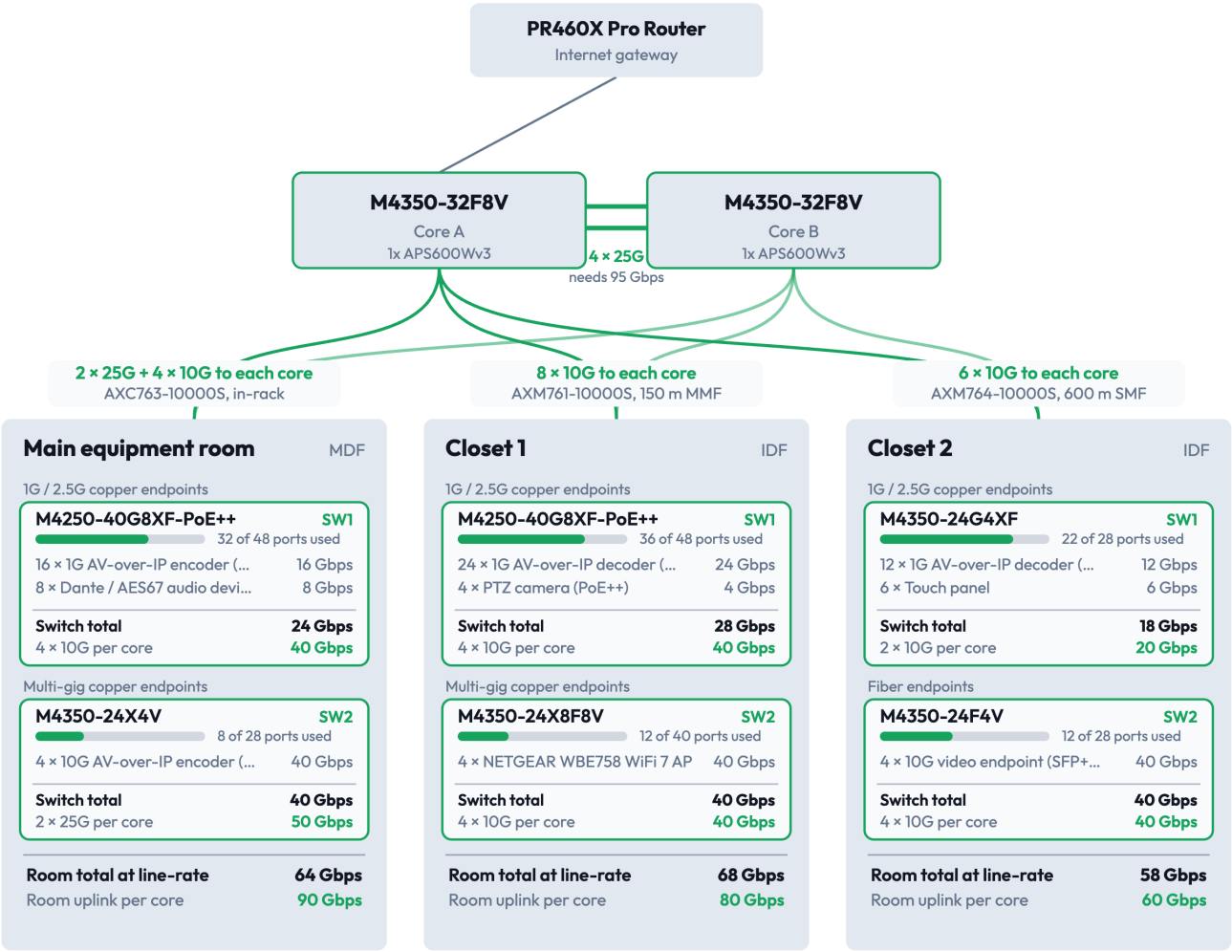
Design summary

This design was validated by the Pro AV Design team on September 30, 2026. Pricing and availability are provided by NETGEAR or your distributor.

ROOMS AND DEVICES

Device	Category	Qty	Link	PoE each
Main equipment room • MDF • 28 devices				
1G AV-over-IP encoder (TX)	Video	16	1G Copper	13 W
10G AV-over-IP encoder (copper)	Video	4	10G Copper	–
Dante / AES67 audio device	Audio	8	1G Copper	13 W
Closet 1 • IDF, 150 m multimode fiber • 32 devices				
1G AV-over-IP decoder (RX)	Video	24	1G Copper	13 W
PTZ camera (PoE++)	Camera	4	1G Copper	60 W
NETGEAR WBE758 WiFi 7 AP	Wireless	4	10G Copper	39 W
Closet 2 • IDF, 600 m single mode fiber • 22 devices				
1G AV-over-IP decoder (RX)	Video	12	1G Copper	13 W
10G video endpoint (SFP+ fiber)	Video	4	10G Fiber	–
Touch panel	Control	6	1G Copper	25 W

Network diagram



Green lines are uplinks, labelled with link count, speed and the optic or cable used. With a redundant core, every switch has its own full-capacity link group to each core, so either core can carry all traffic alone.

Bill of materials

Part number	Description	Qty	Where
SWITCHES			
GSM4248UX-100NAS	M4250-40G8XF-PoE++ switch	2	Main equipment room, Closet 1
GSM4328-100NES	M4350-24G4XF switch	1	Closet 2
XSM4328CV-100NES	M4350-24X4V switch	1	Main equipment room
XSM4328FV-100NES	M4350-24F4V switch	1	Closet 2
XSM4340FV-100NES	M4350-32F8V core switch	2	Main equipment room (core)
XSM4340V-100NES	M4350-24X8F8V switch	1	Closet 1
POWER			
APS350W-100NES	350W modular PSU	2	Main equipment room, Closet 2
APS600W-200NES	600W modular PSU (v2)	1	Closet 2
APS600W-300NES	600W modular PSU (v3)	3	Closet 1, Main equipment room (core)
OPTICS			
AXM761-10000S	10GBASE-SR SFP+, multimode	36	Closet 1, Closet 2
AXM764-10000S	10GBASE-LR LITE SFP+, single mode	24	Closet 2
CABLING			
AXC763-10000S	10G SFP+ DAC 3m, passive copper	9	Main equipment room
To be confirmed	25G direct-attach cable (not in catalog)	8	Main equipment room, Main equipment room (core)
WIRELESS			
WBE758-111NAS	NETGEAR WBE758 WiFi 7 AP	4	Closet 1
GATEWAY			
PR460X-111NAS	PR460X Pro Router	1	Main equipment room

Part numbers are for Americas. Pricing and availability are provided by NETGEAR or your distributor after validation.

Power and rack space

Room	Est. draw	Datasheet max	Heat load	Current	Circuits	UPS size	Rack
Main equipment room	705 W	4,587 W	2,404 BTU/hr	6.4 A at 110 V	1 × 20 A	783 VA+	4.0 U
Closet 1	1,021 W	5,580 W	3,484 BTU/hr	9.3 A at 110 V	1 × 20 A	1,135 VA+	3.0 U
Closet 2	539 W	991 W	1,840 BTU/hr	4.9 A at 110 V	1 × 20 A	599 VA+	2.0 U
Main equipment room (core)	313 W	313 W	1,067 BTU/hr	2.8 A at 110 V	1 × 20 A	348 VA+	2.0 U

POWER SUPPLIES PER SWITCH

Room	Device	Qty	PoE load each	Power supplies	PoE available (1 PSU failed)	Headroom	Est. draw each
Main equipment room	M4250-40G8XF-PoE++	1	312 W	2 of 3 PSUs powered	720 W	57%	429 W
Main equipment room	M4350-24X4V	1	–	1x APS350W	700 W	100%	119 W
Closet 1	M4250-40G8XF-PoE++	1	552 W	2 of 3 PSUs powered	720 W	23%	696 W
Closet 1	M4350-24X8F8V	1	156 W	1x APS600Wv3	290 W	46%	325 W
Closet 2	M4350-24G4XF	1	306 W	1x APS600Wv2	468 W	35%	420 W
Closet 2	M4350-24F4V	1	–	1x APS350W	–	–	119 W
Main equipment room (core)	M4350-32F8V	2	–	1x APS600Wv3	–	–	156 W
Main equipment room	PR460X Pro Router	1	–	Internal PSU	–	–	156 W

Estimated draw = datasheet maximum without PoE + PoE load ÷ 0.9 (PSU efficiency). Circuits assume 80% continuous load. Feed each power supply from a separate circuit or UPS.

Checks

CHECK	Main equipment room: no 25G cable in the catalog for 3 m. Placeholder added.
NOTE	Inter-core link sized for half of all room traffic (active/active cores), and never below the busiest single switch: needs 95 Gbps, provided $4 \times 25G = 100$ Gbps.
CHECK	No 25G inter-core cable in the catalog. Placeholder added.
NOTE	Devices above 30W need Ultra90 PoE++ (802.3bt) ports; only those switches were considered for them.
NOTE	Switches interconnect with LAG uplinks and IGMP Plus (no stacking), which keeps AVB and PTP available.
NOTE	Uplink LAGs use 1, 2, 4 or 8 links to each core, and each core link group carries the switch's full load on its own so one core can fail (sizes from Tool_Settings: 1, 2, 4, 8), so LAG hashing spreads traffic evenly.
NOTE	Redundant power: PoE is sized to the budget that remains if one power supply fails.

UPLINK PLAN

Room	Switch	Units	Uplinks each	Speed	Run	Optic / cable
Main equipment room	M4250-40G8XF-PoE++	1	8	10G	in-rack	AXC763-10000S
Main equipment room	M4350-24X4V	1	4	25G	in-rack	To be confirmed
Closet 1	M4250-40G8XF-PoE++	1	8	10G	MMF, 150 m	AXM761-10000S
Closet 1	M4350-24X8F8V	1	8	10G	MMF, 150 m	AXM761-10000S
Closet 2	M4350-24G4XF	1	4	10G	SMF, 600 m	AXM764-10000S
Closet 2	M4350-24F4V	1	8	10G	SMF, 600 m	AXM764-10000S